
FREMS™
NEUROVASCULAR REHABILITATION

Recentemente la stimolazione elettrica è stata inserita tra i trattamenti consigliati nella cura delle ulcere cutanee croniche

Electrical Stimulation

Consider the use of direct contact (capacitative) electrical stimulation (ES) in the management of recalcitrant Category/Stage II, as well as Category/Stage III and IV pressure ulcers to facilitate wound healing. (Strength of Evidence = A)
EPUAP Final Consensus Treatment, Quick Reference Guide 2009



L'abilità di una ES di risultare efficace nel miglioramento delle ulcere dipende dalle caratteristiche fisiche della corrente ma soprattutto dalle modalità di erogazione del segnale elettrico.

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A partire da queste considerazioni è stata progettata e realizzata



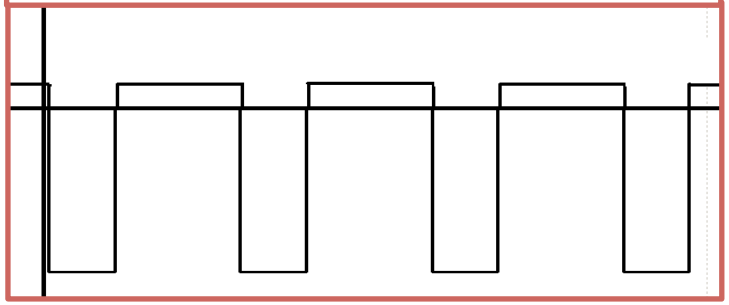
Una neurostimolazione elettrica biocompatibile di nuova generazione che con le sue caratteristiche fisiche di segnale computerizzato e con le evidenze cliniche a supporto si propone come trattamento per il

miglioramento della sintomatologia dolorosa nelle neuropatie sintomatiche diabetiche periferiche, associate o no a vasculopatie

e nelle patologie ulcerative croniche.

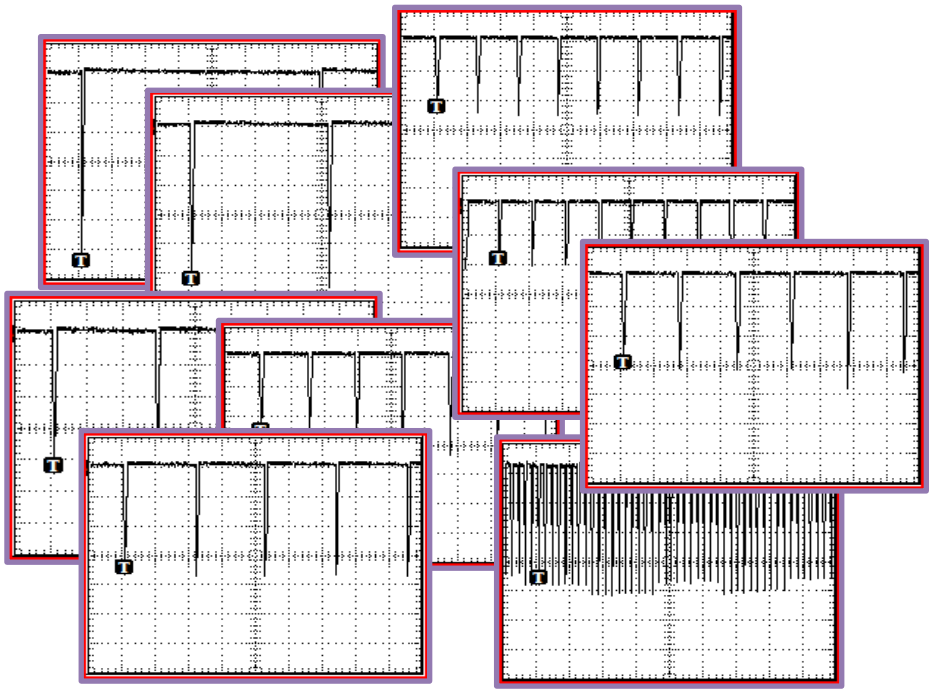
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TENS tradizionale



Segnale periodico, frequenza fissa

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Impulsi bifasi asimmetrici a valore medio nullo;
Segnale computerizzato (multimodulazione di frequenza ampiezza e durata del segnale)

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TECNOLOGIA VALIDATA DA STUDI SCIENTIFICI PUBBLICATI

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Results: 7

Induction of vascular endothelial growth factor release by transcutaneous frequency modulated neural stimulation in diabetic polyneuropathy.
4. Bevilacqua M, Dominguez LJ, Barrella M, Barbagallo M.
J Endocrinol Invest. 2007 Dec;30(11):944-7.
PMID: 18250616 [PubMed - indexed for MEDLINE]
Related citations

Frequency rhythmic electrical modulation system (FREMS) on H-reflex amplitudes in healthy subjects.
5. Barrella M, Toscano R, Goldoni M, Bevilacqua M.
Eura Medicophys. 2007 Mar;43(1):37-47. Epub 2006 Dec 12.
PMID: 17159742 [PubMed - indexed for MEDLINE] Free Article
Related citations

A randomized controlled study on the effect of two different treatments (FREMS AND TENS) in myofascial pain syndrome.
6. Farina S, Casarotto M, Benelle M, Tinazzi M, Fiaschi A, Goldoni M, Smania N.
Eura Medicophys. 2004 Dec;40(4):293-301.
PMID: 16175154 [PubMed] Free Article
Related citations

Effectiveness of frequency-modulated electromagnetic neural stimulation in the treatment of painful diabetic neuropathy.
7. Bosi E, Conti M, Vermigli C, Cazzetta G, Peretti E, Cordoni MC, Galimberti G, Scionti L.
Diabetologia. 2005 May;48(5):817-23. Epub 2005 Apr 15.
PMID: 15834546 [PubMed - indexed for MEDLINE]

MECCANISMI D'AZIONE

Grazie alle sue caratteristiche fisiche è in grado di aumentare la *vasomozione* ed il *flusso sanguigno periferico*,

Recovery of 0,1 Hz microvascular skin blood flow in disautonomic diabetic (type 2) neuropathy by using FREMS
L. Bocchi et al , Medical Engineering and Physics, 2010

di indurre la *modulazione di fattori di crescita angiogentici* (VEGF)

Induction of vascular endothelial growth factor release by transcutaneous frequency modulated neural stimulation in diabetic polyneuropathy.
Bevilacqua M. et al. J Endocrinol Invest. 2007

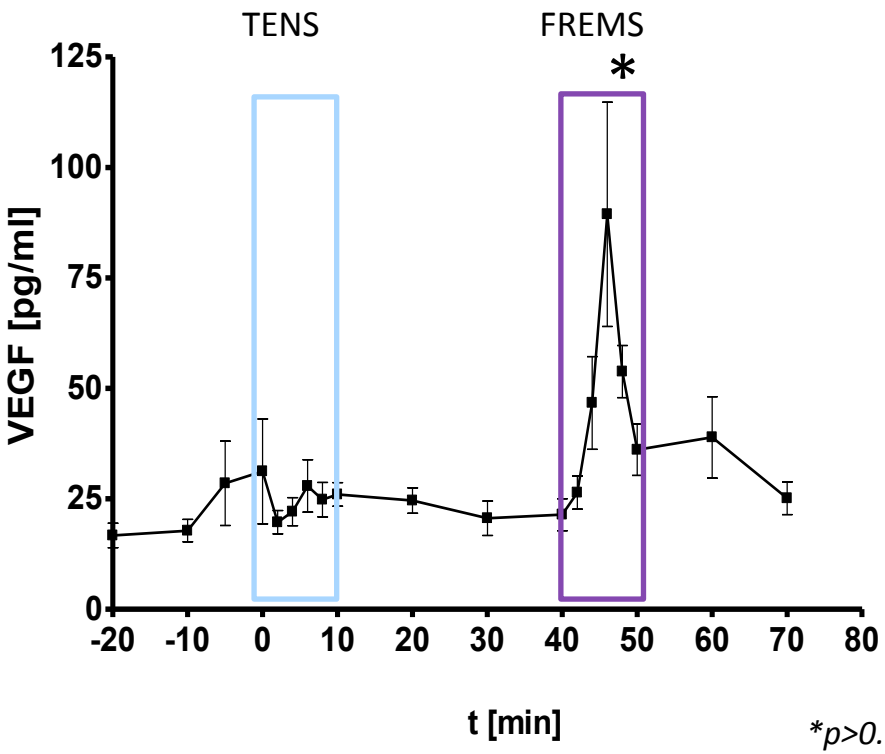
ed il *recupero funzionale del microcircolo* e del sistema nervoso periferico

FREMS enhances cutaneous microvascular flow in patients with diabetic neuropathy
Conti M. et al. J Diabetes Complications, 2008

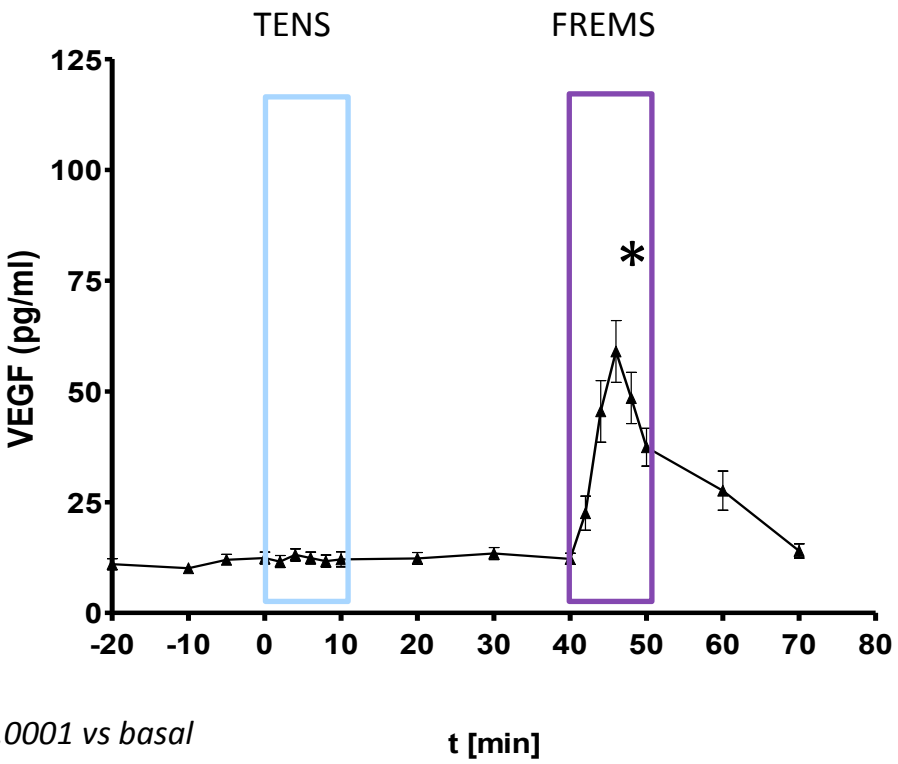
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MODULAZIONE VEGF

Soggetti normali



Soggetti diabetici



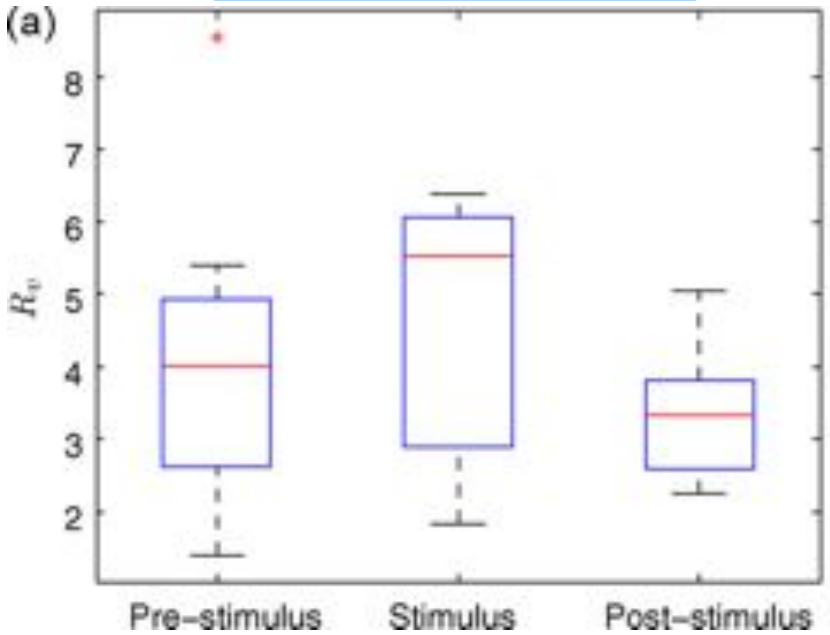
**p>0.0001 vs basal*

Induction of vascular endothelial growth factor release by transcutaneous frequency modulated neural stimulation in diabetic polyneuropathy.
Bevilacqua M. et al. J Endocrinol Invest. 2007

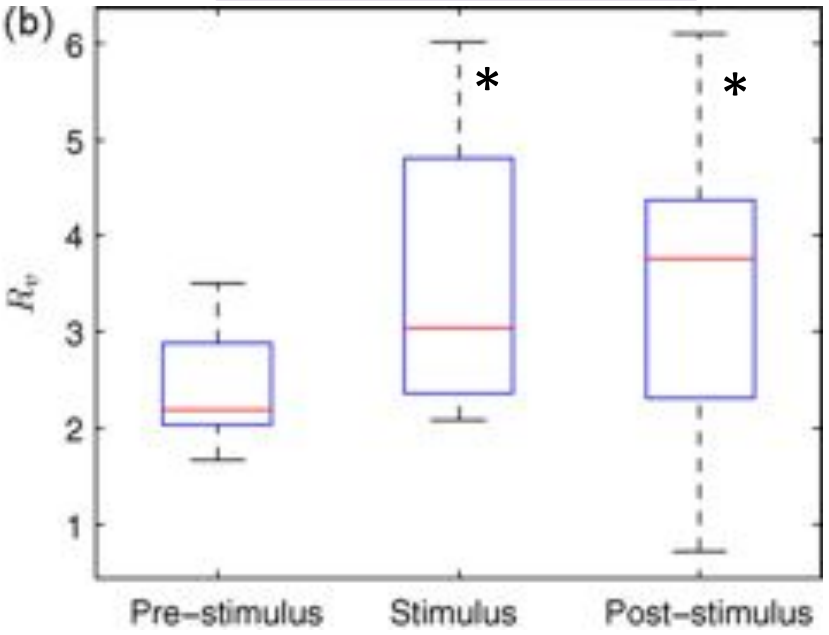
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AUMENTO DEL FLUSSO SANGUIGNO

Gruppo Controllo



Gruppo Diabetici



Il valore di Rv nei pz controllo torna ai valori basali dopo la stimolazione mentre nei pz diabetici rimane statisticamente significativo anche dopo la stimolazione

Recovery of O2 in microvascular skin blood flow in disautonomic diabetic (type 2) neuropathy by using FREMS

L. Bocchi et al , Medical Engineering and Physics, 2010

EFFETTI CLINICI

NEI PAZIENTI CON PATOLOGIE VASCOLARI E NERVOSE PERIFERICHE

Neuropatie diabetiche, Arteriopatie, Ulcere cutanee croniche

- Riduzione del dolore
- Aumento dei Valori Ossimetrici *(in pz con PAD)*
- Riduzione dei tempi di guarigione
- Favorisce la riepitelizzazione

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ESPERIENZA CLINICA

Sono stati effettuati studi clinici su trattamenti specifici e differenziati nei seguenti ambiti

Neuropatia periferica dolorosa diabetica

Arteriopatie periferiche

Ulcere cutanee croniche dell'arto inferiore

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FREMS™ Neuropatia diabetica

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Diabetologia (2005) 48: 817–823
DOI 10.1007/s00125-005-1734-2

ARTICLE

E. Bosi · M. Conti · C. Vermigli · G. Cazzetta · E. Peretti ·
M. C. Cordoni · G. Galimberti · L. Scionti

Effectiveness of frequency-modulated electromagnetic neural stimulation in the treatment of painful diabetic neuropathy

Studio clinico randomizzato doppio cieco

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Risultati

31 pazienti affetti da Neuropatia periferica diabetica dolorosa

10 sedute di FREMS™

820

Table 2 Effect of treatment with FREMS or placebo

	n	Placebo		p value	FREMS		p value
		Pre-treatment	Post-treatment		Pre-treatment	Post-treatment	
VAS daytime pain score	31	31.2±3.9	31.9±4.2	ns	37.1±5.3	26.2±3.9	0.0025
VAS night-time pain score	31	33.3±3.8	30.4±4.2	ns	38.1±5.5	28.5±3.8	0.0197
VPT (V)	31	34.7±1.6	34.2±1.6	ns	35.5±1.6	33.4±1.6	0.0001
Monofilament (out of nine standard areas)	12	5.1±0.9	5.2±0.9	ns	5.8±0.8	4.6±0.9	0.0077
MNCV (m/s)	26	37.2±1.2	37.4±1.4	ns	35.7±1.3	40.5±1.8	0.0019
SNCV (m/s)	15	31.9±2.1	30.2±2.7	ns	30.4±2.9	33.0±2.3	ns
SF36	31	104.4±1.5	105.9±1.5	ns	103.7±1.5	105.6±1.3	ns

VPT vibration perception threshold; *VAS* visual analogue scale; *MNCV* motor nerve conduction velocity; *SNCV* sensory nerve conduction velocity; *SF36* Short Form-36 questionnaire

Bosi et al. Diabetologia (2005) 48: 817–823.

Risultati

Follow up a 4 mesi

FREMS

Effetti duraturi

Table 3 Comparison of results at 4 months with baseline values

	n	Baseline	4-month follow-up	p value
VAS daytime pain score	31	37.0±5.3	25.1±4.2	<0.01
VAS night-time pain score	31	41.1±5.2	26.5±3.9	<0.01
VPT	31	35.6±1.6	31.7±1.8	<0.05
Monofilament	12	5.8±0.8	4.7±0.9	<0.01
MNCV (m/s)	26	35.5±1.2	38.7±1.4	<0.05
SNCV (m/s)	15	26.6±2.8	33.1±3.1	ns
SF36	31	103.6±1.5	107.9±1.2	<0.001
General health	31	4.9±0.3	4.9±0.2	ns
Physical functioning	31	23.1±0.9	25.0±0.7	<0.05
Role limitation due to physical and social functioning	31	6.1±0.3	6.6±0.3	<0.01
Social functioning	31	8.5±0.4	9.2±0.3	<0.05
Bodily pain	31	6.2±0.4	6.8±0.3	<0.05
General mental health	31	37.9±0.7	39.0±0.5	<0.05
Role limitation due to emotional problems	31	3.5±0.2	3.4±0.1	ns
Vitality and health perception	31	13.4±0.3	13.0±0.4	ns

VPT vibration perception threshold; *VAS* visual analogue scale; *MNCV* motor nerve conduction velocity; *SNCV* sensory nerve conduction velocity; *SF36* Short Form-36 questionnaire

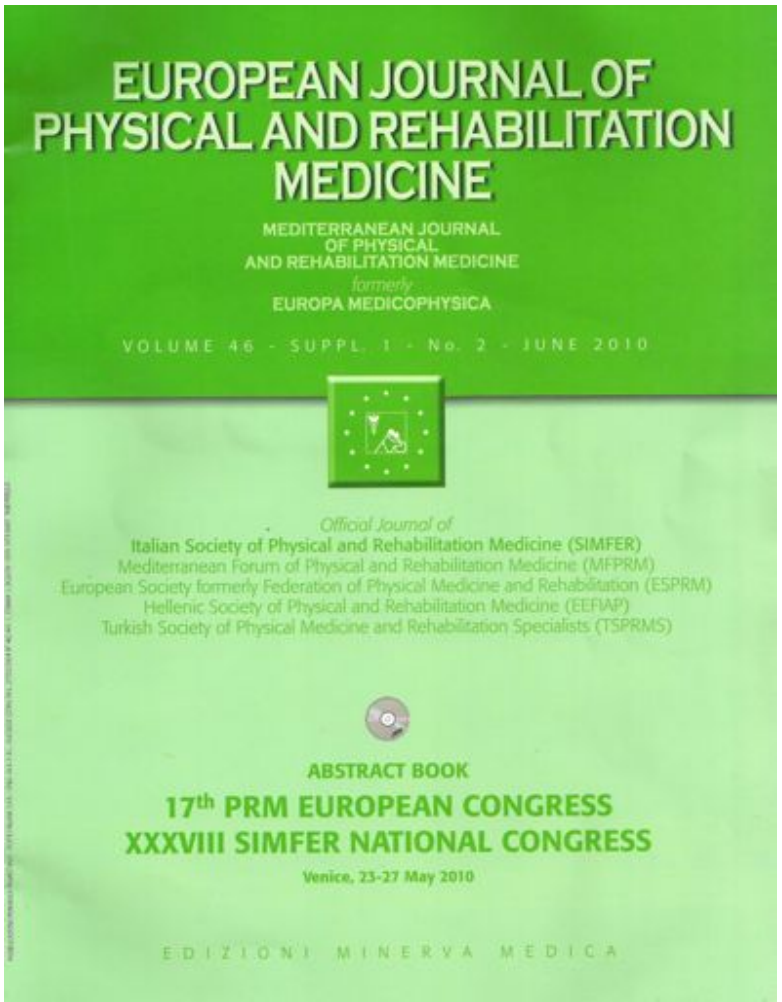
Bosi et al. Diabetologia (2005) 48: 817–823.

FREMS™ Arteriopatie periferiche

Treatment by neuroelectrostimulation (FREMS) of the artheriopaty of lower limb: preliminary study

S. Michelini, M. Cardone, L. Michelotti, G. Moneta, A. Fiorentino, F. Cappellino

San Giovanni battista Hospital, ACISMOM, Rome, Italy



24 pazienti → 2 gruppi (A e B)

Prima

Treadmill test , Laser Doppler

10 Sedute

Dopo

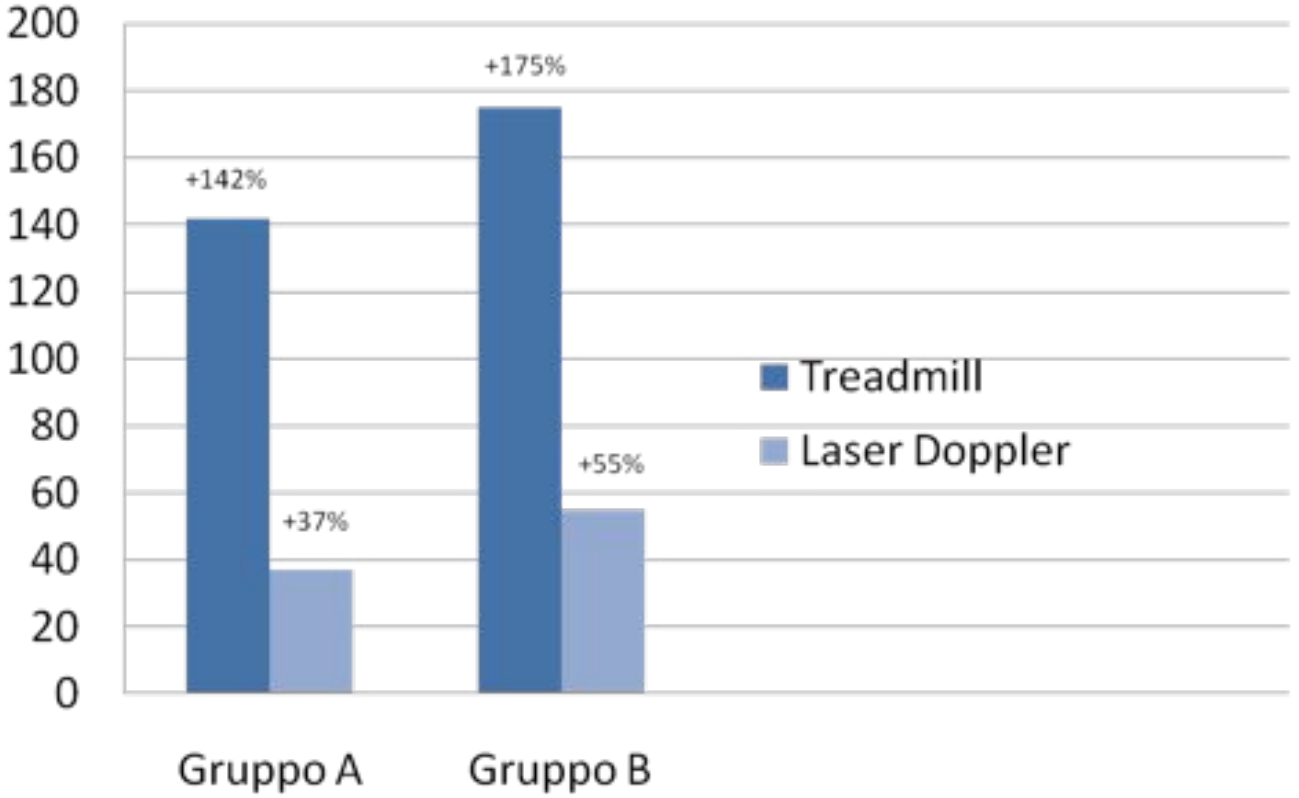
Treadmill test , Laser Doppler

Gruppo A

Massaggi e Kinesiterapia

Gruppo B

Massaggi e Kinesiterapia
+ FREMS



Neuromodulation FREMS in the treatment of diabetic peripheral arterial disease

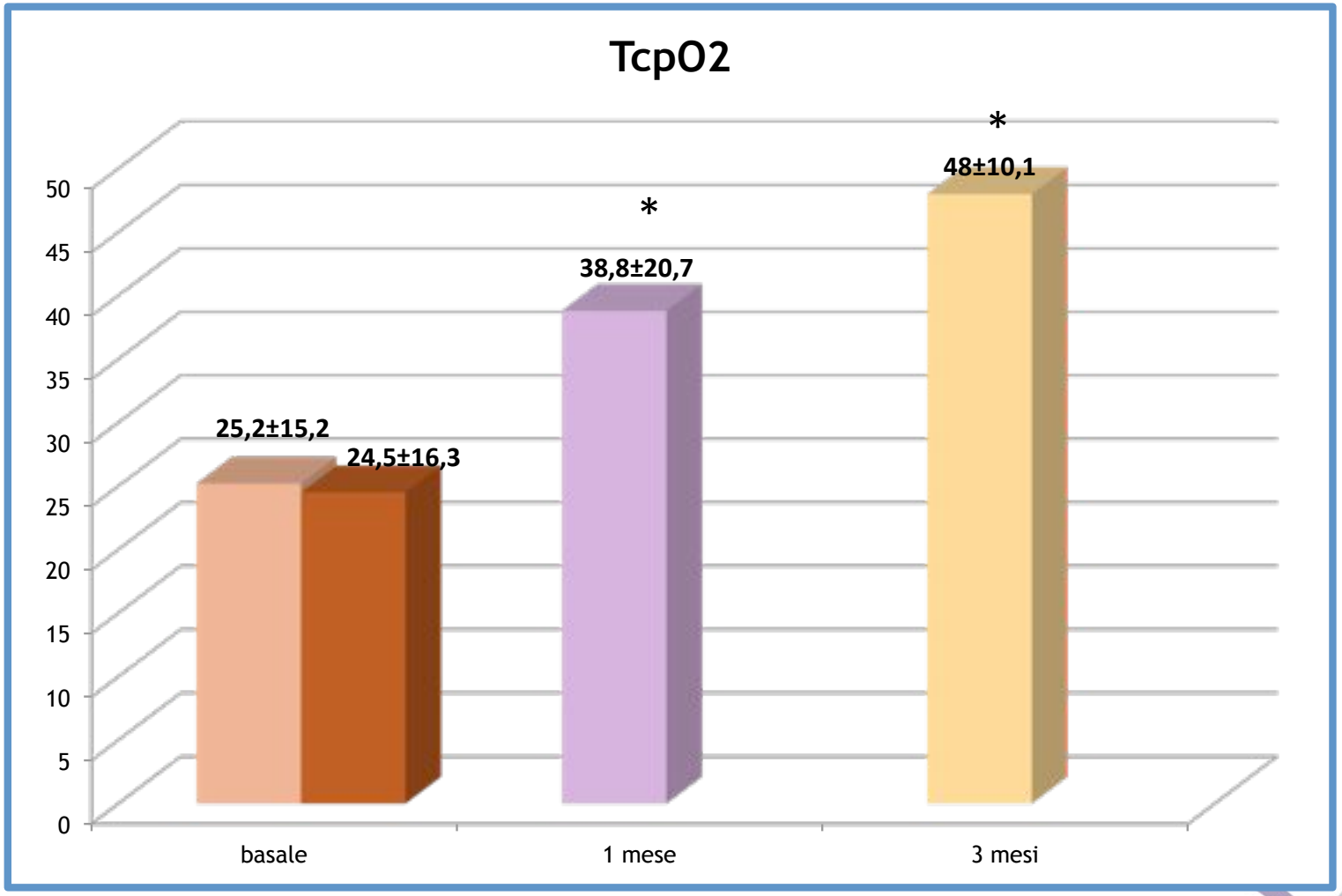
Roberto Da Ros, Cristiano Vitale[$], Roberta Assaloni, Antonio Ceriello[°]
Foot and Ankle Clinic, Abano Terme, [$] University of Udine, [°] Università di Warwick,

Abstracts of the 42nd EASD Annual Meeting, Copenhagen–Malmoe, Denmark–Sweden, 14–17 September 2006

Diabetologia
Journal of the European Association for the Study of Diabetes (EASD)

Risultati

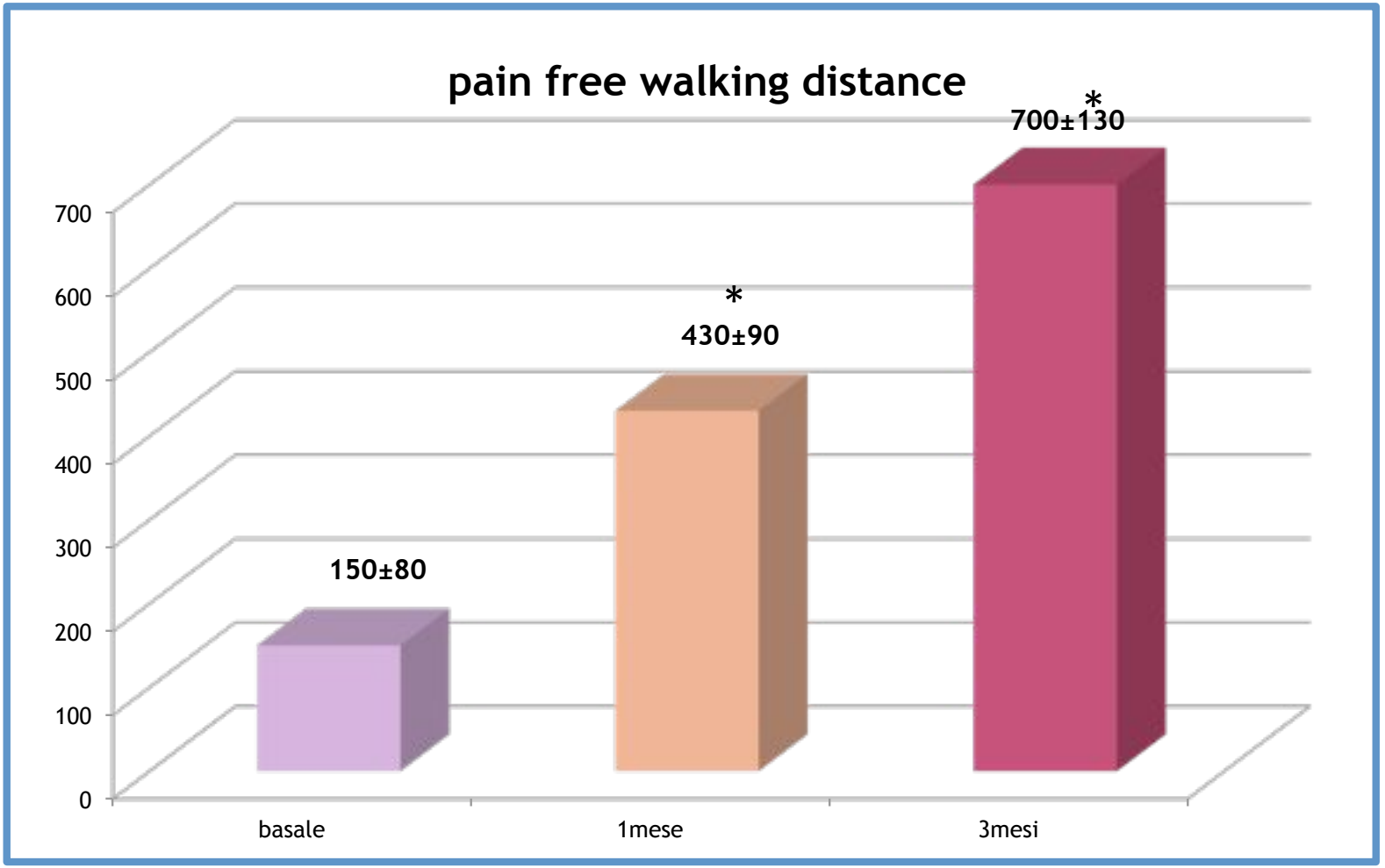
18 pazienti
10 sedute di FREMS, follow up a 1 e 3 mesi



Da Ros, et al, Università di Udine, University of Warwick, Foot and Ankle Clinic, Abano Terme – EASD 2006

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Risultati



FREMS
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FREMS
Ulcere cutanee croniche

FREMS
NEUROVASCULAR REHABILITATION

Arch Dermatol Res (2008) 300:377–383
DOI 10.1007/s00403-008-0875-9

ORIGINAL PAPER

Frequency rhythmic electrical modulation system in the treatment of chronic painful leg ulcers

Aleksandar Janković · Ivana Binić

Received: 27 July 2007 / Revised: 23 June 2008 / Accepted: 26 June 2008 / Published online: 16 July 2008
© Springer-Verlag 2008

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Risultati

43 ulcere

pz FREMS *(15 sedute FREMS + tratt.trad.)* vs pz CONTROLLO *(tratt. trad.)*

Diminuzione della superficie dell'ulcera

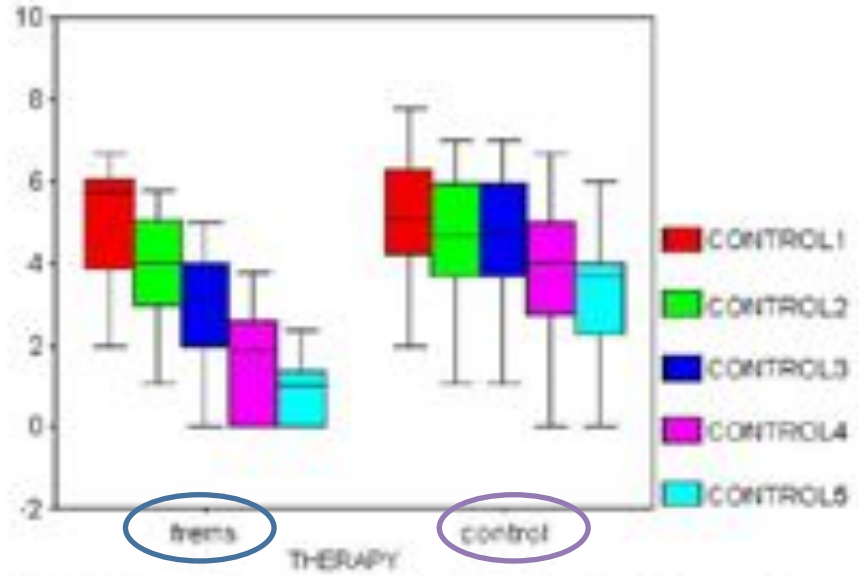


Fig. 1 Decrease of mean surface leg ulcer FREMS and control

Diminuzione del Dolore a fine trattamento

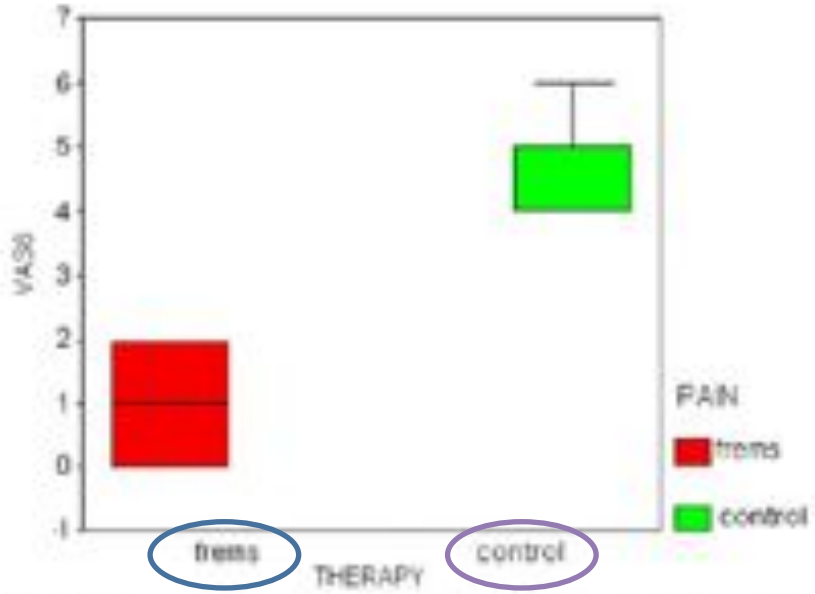


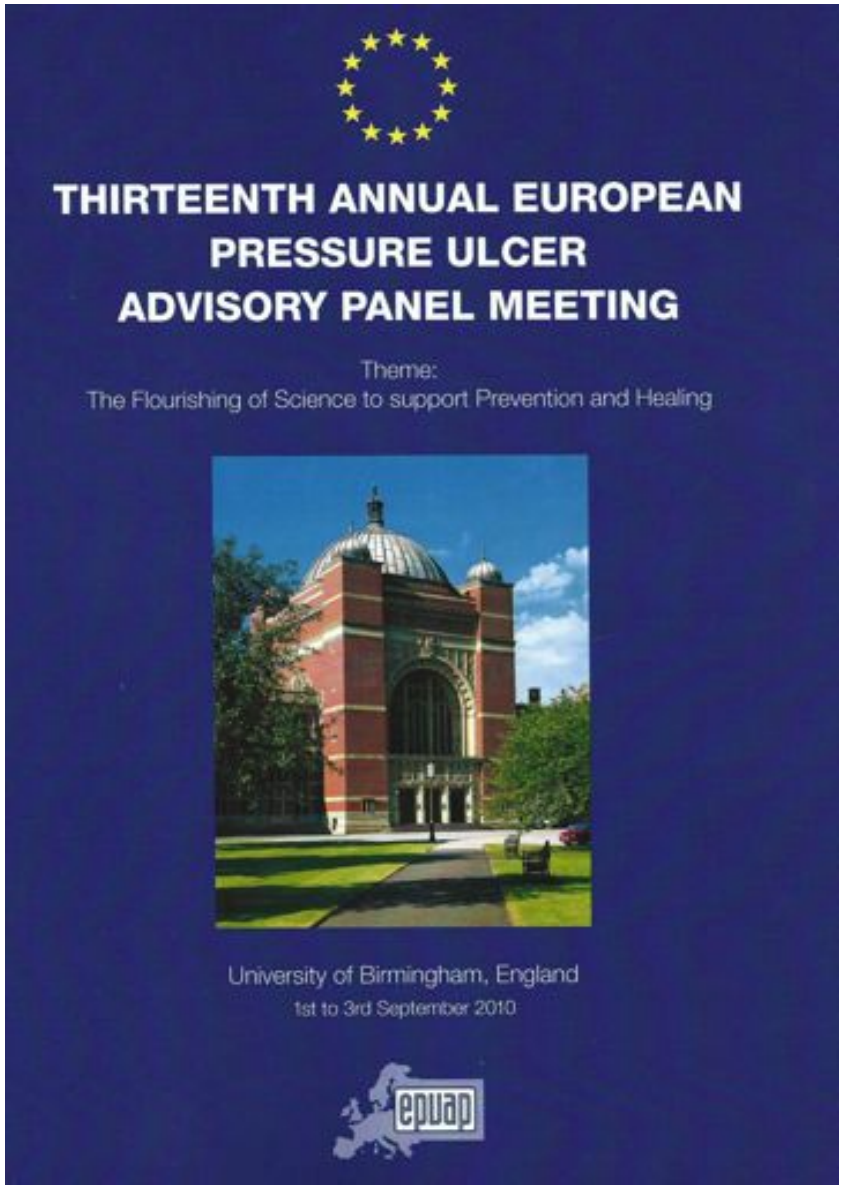
Fig. 2 Decrease of mean VAS score FREMS and control at the end of treatment

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Randomized controlled clinical study on the application of the treatment FREMS™ in the management of chronic ulcer of lower limb

Federica Neri,[1*], Olezzi Daniela [1*], Samaras Anna [1*] Gianetti Alberto [1*] Cristina Magnoni [1*]

[1*]Dermatological Clinic, University of Modena and Reggio Emilia, cristina.magnoni@unimore.it



STUDIO CLINICO ULCERE CUTANEE CRONICHE

UNIVERSITÀ DI MODENA E REGGIO EMILIA

La percentuale di guarigione nel gruppo FREMS è sempre superiore al gruppo di controllo (p<0.0001 paired student t-test); in particolare la percentuale di guarigione nel gruppo FREMS è significativamente superiore al controllo al termine del 2° e del 3° ciclo di terapia, dove si vede maggiormente l'efficacia della terapia.

